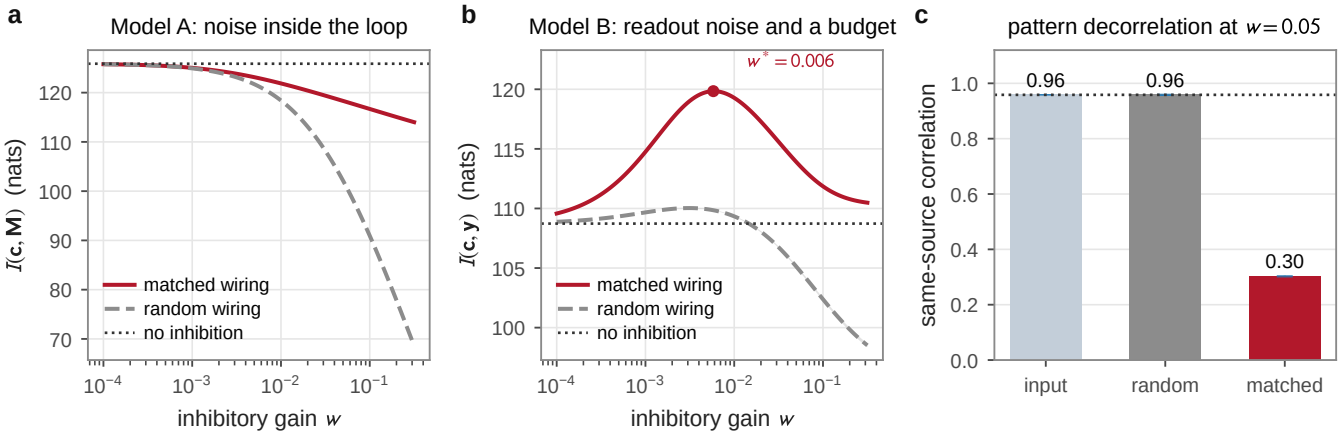
a
Model A: noise inside the loop
$\mathcal{I}(\mathbf{c}, \mathbf{M})$ (nats)
120
110
100
90
80
70
matched wiring
random wiring
no inhibition
10^{-4}
10^{-3}
10^{-2}
10^{-1}
inhibitory gain w
b
Model B: readout noise and a budget
$w^* = 0.006$
$\mathcal{I}(\mathbf{c}, \mathbf{y})$ (nats)
120
115
110
105
100
matched wiring
random wiring
no inhibition
10^{-4}
10^{-3}
10^{-2}
10^{-1}
inhibitory gain w
c
pattern decorrelation at $w = 0.05$
same-source correlation
1.0
0.8
0.6
0.4
0.2
0.0
0.96
0.96
0.30
input
random
matched